Every step shown | **Tagged** by lesson + CLO | **File** 8 / 8

ANS

Chapter 3 full worked answers

الإجابات الكاملة بالخطوات: افتحه بعد حل ملف الأسئلة

Each answer shows the rule, the substitution, the number, and the interpretation sentence a marker looks for. The `why` lines name the trap the question was built around. All values were computed by code from the question data, not typed by hand.

Part A · Concept and interpretation (multiple choice)

Each item needs a reason, not a recalled fact. Give the letter *and* one line of justification, as in a written exam.

A1 3-1 CLO1-A

(b)

The question defines the population as all 2025 emergency patients, and every one of them was measured. A measure of the whole population is a **parameter** (μ). Option (a) is the classic trap: being a mean doesn't make it a statistic.

A2 3-1 CLO3-C

(b)

The mean is far above the median, so the distribution is **right-skewed** by a few large salaries (the Example 3-11 pattern). At least half earn OMR 820 or less, which rules out (c). In right-skewed data the mode is usually the smallest of the three, which rules out (d).

A3 3-1 CLO2-A

(c)

The mean needs $\Sigma f \cdot X_m$, and an **open-ended class has no midpoint**. The median class and modal class need only frequencies and cumulative frequencies, so both can still be found.

A4 3-1 CLO3-B

(b)

Mean < median < mode: the mean has been pulled toward a long **left** tail, so the distribution is negatively skewed.

A5 3-1 CLO2-A

(d)

12 occurs 20 times, so it is the mode. "No mode" means no value repeats, which is the opposite situation. Every deviation is 0, so s = 0.

A6 3-2 CLO3-C

(a)

With **equal means**, the smaller SD means less variation. The CV would give the same ranking: 0.72% vs 1.64%. The CV is essential only when the means or units differ, so (d) is wrong here.

A7 3-2 CLO1-A

(b)

The variance is an average of **squared** deviations, so it is never negative, however the values sit relative to the mean.

A8 3-2 CLO2-A

(b)

Shape unknown, so use Chebyshev: $1 - 1/2.5^2 = 1 - 0.16 = 0.84 =$ **84%**. 68% and 95% are empirical-rule values and need a bell shape.

A9 3-2 CLO2-A

(b)

70 = 60 + 2(5). About 95% lie within 50–70, so 5% lie outside, split equally between the two tails: **2.5%** above 70.

A10 3-2 CLO3-C

(b)

Adding a constant shifts every salary equally, so the spread (SD) is unchanged. The mean rises by 50, so CV = $s/\bar{x}$ gets **smaller**.

A11 3-2 CLO2-A

(c)

Multiplying by c multiplies the SD by c and the **variance by c^2**: $6.5 \times 9 = 58.5$.

A12 3-3 CLO3-C

(b)

A z-score counts **standard deviations** from the mean; the sign says below. It says nothing about percentages correct (a) or marks (d).

A13 3-3 CLO3-C

(b)

A **percentile** is a relative position, not a percentage of marks. This is Dr. Amani's percentile vs percentage distinction.

A14 3-3 CLO1-A

(d)

Median = P_{50} = Q_2 = D_5. P_5 is the 5th percentile.

A15 3-3 CLO3-C

(c)

The SD is built from squared distances from the mean, so one extreme value inflates it dramatically. The median, Q_1 and IQR depend on positions and move very little.

A16 3-4 CLO3-B

(c)

Median right of centre and left whisker longer both point to a long **left** tail: negatively skewed.

A17 3-4 CLO1-A

(b)

EDA uses the **median** and the **IQR**, because neither is distorted by the outliers EDA is trying to find.

A18 3-4 CLO3-B

(b)

Each of the four pieces holds **25%** of the data. A longer piece means the same number of values spread over a wider interval, not more values.

Part B · Short exam problems by lesson

Inverse problems, corrections, comparisons and interpretation, the way quiz and midterm questions are actually built.

B1 3-1 CLO2-A

$\Sigma x = n \cdot \bar{x} = 6 \times 14.5 = 87$. Sixth value = 87 − 75 = **12**.

Ordered: 9, 12, 12, 16, 18, 20. n = 6 (even), so median = (12 + 16)/2 = **14**. Mode = **12**.

Key idea: $\bar{x} = \Sigma x / n$ can be run backwards to give $\Sigma x = n \cdot \bar{x}$. Exam favourite: a missing value or a correction.

B2 3-1 CLO2-A

(a) Recorded total = 25 × 68 = 1700. Corrected total = 1700 − 48 + 84 = 1736. Correct mean = 1736/25 = **69.44**.

(b) **Not necessarily.** The median is the 13th ordered value. Moving one score from 48 to 84 changes it only if that score crosses the middle position, and we can't tell without the data. The mean always changes, because it uses every value.

B3 3-1 CLO2-A · CLO3-C

No. The sections are different sizes, so their means must be **weighted** by the number of students.

$$\bar{x} = \frac{32(71) + 18(79)}{32 + 18} = \frac{2272 + 1422}{50} = \mathbf{73.88}$$

The overall mean sits closer to 71 because the larger section carries more weight.

B4 3-1 CLO2-A

Weighted points so far: 0.15(80) + 0.10(90) + 0.05(100) + 0.25(72) = 12 + 9 + 5 + 18 = 44.

(a) An A needs 0.45F ≥ 90 − 44 = 46, so F ≥ 102.22. That is impossible: **no**. The best possible total is 44 + 45 = 89.

(b) 0.45F ≥ 86 − 44 = 42, so F ≥ 93.33: with whole-number marks the student needs at least **94** on the final.

B5 3-1 CLO2-A · CLO3-B

Mean (each value weighted by its frequency): (0·5 + 1·9 + 2·14 + 3·8 + 4·4)/40 = 77/40 = **1.925**.

Median: n = 40, so average the 20th and 21st values. Cumulative counts: 5, 14, 28, ..., so both lie in the "2" group: median = **2**. Mode = **2** (14 families).

Mean (1.925) < median = mode (2): **slightly negatively skewed**.

B6 3-1 CLO3-C

(a) **Mode**: nominal data, so no ordering or arithmetic is meaningful. (b) **Median**: ordinal data, which can be ordered but not averaged. (c) **Median**: ratio data, but right-skewed by outliers that would inflate the mean. (d) **Mean**: ratio data, not skewed; the mean uses all values and varies least across samples.

B7 3-1 CLO2-A · CLO3-C

(a) n = 15, Σx = 737, so $\bar{x}$ = **49.133**. Median = 8th value = **49**. Mode = **52**. MR = (32 + 78)/2 = **55**.

(b) The mean rises by 20/15 = **1.33**, to 50.467. The midrange rises by 20/2 = **10**, to 65. The **median and mode do not change**: they depend on positions and repeats, not on how far the largest value is.

B8 3-2 CLO2-A

$$s^2 = \frac{12(25{,}140) - 540^2}{12(11)} = \frac{301{,}680 - 291{,}600}{132} = \frac{10{,}080}{132} = 76.3636$$

s = **8.739**; $\bar{x}$ = 540/12 = 45; CV = 8.739/45 × 100 = **19.42%**.

The raw data are not needed: the shortcut formula works directly from n, ΣX and ΣX².

B9 3-2 CLO2-A

The student divided by n. This is a **sample**, so the divisor is n − 1 = 3.

$s^2 = 34/3 =$ **11.333** $\quad s =$ **3.367**

Dividing by n underestimates the population variance; that is the reason for n − 1.

B10 3-2 CLO2-A · CLO3-C

The units differ (OMR vs °C), so the raw SDs can't be compared. Use the CV:

price: 0.008/0.229 × 100 = 3.49% temperature: 2.6/34 × 100 = 7.65%

Temperature was relatively more variable; the petrol price was more consistent.

B11 3-2 CLO2-A

Set 1 − 1/k² = 0.96, so 1/k² = 0.04, k² = 25 and k = 5.

80 ± 5(3) → **65 to 95**

Chebyshev run backwards: from the percentage to k, then to the interval.

B12 3-2 CLO2-A · CLO3-C

Chebyshev applies to intervals **symmetric** about the mean. The interval 40–65 is 2 SD below and 3 SD above. The symmetric interval it is guaranteed to contain is 40–60 (k = 2), so at least **75%** lie between 40 and 65. The student's 84% isn't guaranteed (and 3 SD would give 88.89%, not 84%).

With an asymmetric interval, use the **smaller** k.

B13 3-2 CLO2-A

(a) 95% lies within ±2 SD, so the mean is the midpoint: (18 + 42)/2 = **30**. Then 4s = 42 − 18 = 24, so s = **6**.

(b) 36 = 30 + 1·6. About 68% lie within ±1 SD, so 32% lie outside, half above: **16%**.

(c) 24 is −1 SD and 42 is +2 SD. Half of 68% (34%) plus half of 95% (47.5%) = **81.5%**.

B14 3-2 CLO2-A · CLO3-C

For y = 5x + 4: centre ×5 + 4, spread ×5, variance ×25.

Mean = 5(12.4) + 4 = **66** · median = 5(13) + 4 = **69** · SD = 5(3.1) = **15.5** · IQR = 5(4.5) = **22.5** · variance = 15.5^2 = **240.25**.

CV before = 3.1/12.4 = 25%; after = 15.5/66 = 23.48%. It **changes**: multiplying alone would leave it unchanged, but the added constant raises the mean without changing the SD.

B15 3-2 CLO2-A

Deviations from the mean always sum to zero: −3 + 5 − 1 + d + 2 = 0, so d = **-3**.

$$s^2 = \frac{9 + 25 + 1 + 9 + 4}{5 - 1} = \frac{48}{4} = \mathbf{12}$$

B16 3-3 CLO2-A · CLO3-C

(a) X = x̄ + z·s = 64 + 1.75(8) = **78**. (b) z = (75 − 64)/8 = 1.375 < 1.75: **no**, he is 3 marks short.

B17 3-3 CLO3-C

z_A = (52 − 55)/2 = -1.5 and z_B = (49 − 50)/0.8 = -1.25.

For running times **lower is better**, so the more negative z is the stronger performance: **Runner A**. A has the slower raw time but is 1.5 SD faster than his group, against 1.25 SD for B.

Trap: picking B because 49 < 52, or picking the larger z without checking whether a high value is good.

B18 3-3 CLO2-A

(a) 9 values are below 30: (9 + 0.5)/16 × 100 = 59.375 ≈ **59th percentile**.

(b) P_{25}: c = 16·25/100 = 4, whole, so (4th + 5th)/2 = (20 + 21)/2 = **20.5**. P_{90}: c = 14.4, rounds up to 15, so the 15th value = **41**.

(c) Q_2 = (26 + 28)/2 = **27**; Q_1 = median of the lower 8 = (20 + 21)/2 = **20.5**; Q_3 = median of the upper 8 = (33 + 35)/2 = **34**. IQR = 13.5; fences 0.25 and 54.25. The extremes 12 and 47 are inside: **no outliers**. Q_1 equals P_{25}, as it should.

(d) $D_3 = P_{30}$: c = 4.8, rounds up to 5, so **21**. $D_7 = P_{70}$: c = 11.2, rounds up to 12, so **33**.

B19 3-3 CLO2-A

(r + 0.5)/40 × 100 = 81.25, so r + 0.5 = 32.5 and r = **32 below**. Above: 40 − 32 − 1 (Maryam herself) = **7**.

B20 3-3 CLO2-A · CLO3-C

Median = (24 + 26)/2 = 25. Q_1 = median of 14 18 20 22 24 = 20. Q_3 = median of 26 28 30 31 45 = 30. IQR = 10 and 1.5·IQR = 15.

Fences: 5 and 45. 45 **equals** the upper fence, and an outlier must be strictly greater, so **45 is not an outlier**.

B21 3-4 CLO3-B · CLO3-C

(a) The median is nearer Q_1 (15 − 11 = 4) than Q_3 (22 − 15 = 7), and the right whisker (27) is much longer than the left (7): **positively skewed**.

(b) IQR = 11, 1.5·IQR = 16.5, upper fence = 38.5. Since 49 > 38.5, **49 is an outlier**.

(c) Half the patients are at or above the median: about **60**. (d) From Q_1 to the maximum is three quarters of the data: **75%**.

B22 3-4 CLO3-C

Centre: Muscat's median (12) is higher than Sohar's (9). **Spread:** Sohar's IQR (11) is larger than Muscat's (7). **Range:** Sohar 25 vs Muscat 18, so Sohar varies more. **Shape:** Muscat is roughly symmetric (median 4 above Q_1 and 3 below Q_3; whiskers 5 and 6). Sohar is **positively skewed** (median only 2 above Q_1; right whisker 12 vs left 2).

Sohar has the shorter typical wait but the longer, less predictable worst cases.

B23 3-4 CLO2-A · CLO3-B

n = 15, median = 8th = **49**. Lower seven 32 35 38 **40** 41 41 46, so Q_1 = **40**. Upper seven 52 52 52 **57** 60 64 78, so Q_3 = **57**.

Five-number summary: 32, 40, 49, 57, 78. IQR = 17; fences 14.5 and 82.5, so 78 is not an outlier.

Shape: the median is almost centred in the box (9 vs 8), but the right whisker (21) is much longer than the left (8): **slightly positively skewed**. The tail comes from 64 and 78.

B24 3-4 CLO3-B · CLO3-C

(a) A: **30, 56, 70, 76, 84** · B: **50, 58, 64, 70, 78**.

(b) **Section A** (70 vs 64). (c) IQR A = 20, IQR B = 12, so **Section A**.

(d) A: the median is right of the box centre (14 above Q_1, 6 below Q_3) and the left whisker (26) is far longer than the right (8): **negatively skewed**. B: median centred (6 and 6), whiskers equal (8 and 8): **symmetric**.

(e) A's median is 70 > 65, so at least half of section A scored above 65. In B the median is 64, so fewer than half did: **Section A**.

Part C · Long structured questions

Midterm-style, multi-part, mixing lessons. Marks shown in brackets.

C1 3-1 3-2 3-3 3-4 CLO2-A · CLO3-B · CLO3-C **[18 marks]**

1. Ordered: 18, 21, 22, 23, 25, 26, 27, 27, 29, 31, 35, 40. $\bar{x}$ = 324/12 = **27**; median = (26 + 27)/2 = **26.5**; mode = **27**; MR = (18 + 40)/2 = **29**.

2. $\Sigma X = 324$, $\Sigma X^2 = 9164$. $s^2 = [12(9164) - 324^2]/(12 \cdot 11) = (109968 - 104976)/132 = 37.8182$, so s = **6.15**. CV = 6.15/27 × 100 = **22.78%**.

3. Lower six 18 21 22 | 23 25 26, so Q_1 = (22 + 23)/2 = **22.5**. Q_2 = 26.5. Upper six 27 27 29 | 31 35 40, so Q_3 = (29 + 31)/2 = **30**. IQR = **7.5**; fences 11.25 and 41.25, so **no outliers** (40 < 41.25).

4. 18, 22.5, 26.5, 30, 40. The median sits 4 above Q_1 and 3.5 below Q_3 (near the centre), but the right whisker (10) is longer than the left (4.5): mildly **positively skewed**. This agrees with part 1: mean (27) > median (26.5).

5. At least **75%**. The interval is 14.7 to 39.3, and 11 of 12 days lie inside: **91.7%**. Chebyshev's bound is a *minimum* and is usually beaten.

6. z = (40 − 27)/6.15 = **2.11**. The busiest day was about 2.1 standard deviations above the mean number of patients.

C2 3-1 3-2 CLO2-A · CLO3-C **[12 marks]**

1. Midpoints 24.5, 34.5, 44.5, 54.5, 64.5. $\Sigma f \cdot X_m = 2205$, $\Sigma f \cdot X_m^2 = 103432.5$.

$$\bar{x} = 2205/50 = 44.1 \qquad s^2 = \frac{50(103432.5) - 2205^2}{50(49)} = 126.367 \qquad s = 11.241$$

$\bar{x}$ = 44.1 years, s = 11.24 years, CV = **25.49%**.

2. Modal class **40–49** (f = 18). N/2 = 25; CF = 5, 17, 35, 45, 50, and 35 is the first CF above 25, so the median class is **40–49**.

3. The mean, s and CV **could not** be found: the open-ended class has no midpoint. The modal class and median class **could**: they use only frequencies, and neither falls in the last class.

C3 3-2 CLO2-A · CLO3-C **[12 marks]**

1. $CV_A = 4/502 = 0.8\%$ and $CV_B = 2.5/498 = 0.5\%$: **B** is more consistent (the means are close, so the SDs agree too).

2. k = (510 − 502)/4 = 2, so at least **75%**.

3. 490 and 514 are 502 ± 3(4), so k = 3. At most 1/9 lie outside: 2000/9 = 222.2, so at most **222** bottles. Round **down** for "at most".

4. 493 and 503 are 498 ± 2(2.5): about **95%**. 495.5 = 498 − 1(2.5); 68% lie within ±1 SD, so half of the remaining 32% is below: about **16%**.

C4 3-2 3-3 CLO2-A · CLO3-C **[12 marks]**

1. 9 values are below 35: (9.5/20) × 100 = **47.5th** percentile. P_{80}: c = 20·80/100 = 16, whole, so (16th + 17th)/2 = (41 + 43)/2 = **42**.

2. z = (41 − 34.8)/8.121 = **0.76**.

3. New mean = 2(34.8) + 3 = **72.6**; new SD = 2(8.121) = **16.243**. The +3 does not affect the SD.

4. Neither changes. New z = (85 − 72.6)/16.243 = 0.76: the shift cancels in the numerator and the ×2 cancels between numerator and denominator. The order of the students is unchanged, so every percentile rank is unchanged too. Position measures don't change under a positive linear transformation.

C5 3-1 3-2 3-3 3-4 CLO3-B · CLO3-C **[14 marks]**

1. Al Khuwair: mean 38.2 well above median 31.5; the right whisker (49) is much longer than the left (12): **positively skewed**. Al Amerat: mean ≈ median (29.4 vs 29.0), the median is nearly centred in the box (4 above Q_1, 5 below Q_3) and the whiskers are equal (10 and 10): **roughly symmetric**.

2. CV: Al Khuwair 14.6/38.2 = 38.2%; Al Amerat 6.1/29.4 = 20.7%. **Al Amerat** is more consistent.

3. Al Khuwair: IQR 23, fences 24 − 34.5 = −10.5 and 47 + 34.5 = 81.5, so **96 is an outlier**. Al Amerat: IQR 9, fences 11.5 and 47.5, so **no outliers**.

4. Al Khuwair: the **median**, because the data are skewed and contain an outlier that inflates the mean. Al Amerat: the **mean** (or either), because the data are roughly symmetric with no outliers.

5. 29.4 ± 2(6.1) = 17.2 to 41.6, so k = 2 and at least 75%: 0.75 × 40 = **30 homes**.

Part D · True / false with justification, and find the mistake

A one-word answer earns nothing. The justification is the mark.

D1 3-2 CLO1-A · CLO3-C

True. s = 0 means $\Sigma(X - \bar{x})^2 = 0$, so every deviation is 0 and every value equals the mean.

D2 3-1 CLO1-A · CLO3-C

False. For even n it is the mean of the two middle values: the median of 2, 4, 6, 8 is 5.

D3 3-2 CLO1-A · CLO3-C

True. The SD is unchanged but the mean shifts, so $s/\bar{x}$ changes.

D4 3-2 CLO1-A · CLO3-C

False. It guarantees at least 75%; the actual percentage is often higher.

D5 3-4 CLO1-A · CLO3-C

False. The whiskers can still differ: 1, 10, 11, 12, 13, 14, 15 has $Q_1 = 10$, $Q_2 = 12$, $Q_3 = 14$ (equal gaps of 2), but the left whisker is 9 and the right is 1, so it is left-skewed.

D6 3-3 CLO1-A · CLO3-C

True. $Q_1 \geq \min$ and $Q_3 \leq \max$, so $Q_3 - Q_1 \leq \max - \min$.

D7 3-1 CLO1-A · CLO3-C

False. The long left tail pulls the mean below the median.

D8 3-2 CLO1-A · CLO3-C

False. The second set is the first minus 4. Shifting does not change the spread, so the variances are equal (0.25).

D9 3-1 CLO2-A

The data were not **ordered**. Ordered: 2, 4, 6, 7, 9, so the median is **6**.

D10 3-3 CLO2-A

c = 7 is a **whole number**, so step 3B applies: P_{35} is **the mean of the 7th and 8th values**. Rounding up and taking a single value is step 3A, used only when c is a decimal.

D11 3-3 CLO2-A

The student counted the values *up to and including* 15 (four of them). The formula counts values strictly **below** X: only 10 and 12, so 2. Correct: (2 + 0.5)/8 × 100 = **31.25th percentile**.

D12 3-2 CLO3-C

The means differ hugely, so comparing raw SDs is invalid. CV_A = 30% and CV_B = 10%: **A is relatively more variable**.

Part E · Dr. Ahmed's review self-study exercises

The three exercises that close Dr. Ahmed's deck, fully solved in the answers file.

E1 3-2 CLO2-A · CLO3-C

Variance 9, so s = 3. The interval is symmetric: 26 ± 14, so k = 14/3 = 4.667. Chebyshev: at least $1 - 1/k^2 = 1 - 9/196 = 95.41\%$ lie *inside*.

So at most 9/196 = **4.59%** lie outside: that is the maximum percentage served.

Two traps: taking s = 9 (it is the variance), and answering with the "inside" percentage.

E2 3-1 3-2 3-3 CLO2-A **[14 marks]**

(a) 6.7 and 7.0 each occur 3 times: **bimodal, 6.7 and 7.0**. Median = (15th + 16th)/2 = (7.0 + 7.0)/2 = **7**. $\bar{x}$ = 208.3/30 = **6.943**. s = **0.875** (sample).

(b) Q_1 = median of the lower 15 = 8th value = 6.4; Q_3 = 23rd value = 7.7. Five-number summary: **5.4, 6.4, 7, 7.7, 8.7**; IQR = **1.3**.

(c) $1 - 1/k^2 = 0.64$, so $k^2 = 1/0.36$ and $k = 5/3 \approx 1.667$. Interval: 6.943 ± 1.667(0.875) = **5.485 to 8.401**.

(d) Only 5.4 and 8.7 fall outside, so 28 of 30 = **93.33%**, well above the guaranteed 64%.

E3 3-1 CLO3-B · CLO3-C

The decisive comparison is mean vs median: 51.3 < 52, so the mean is pulled toward a left tail: **negatively (left) skewed**.

The mode (46.5) does not fit the textbook ordering mean < median < mode. With only ten values the mode is unstable (any repeated value becomes the mode), so base the conclusion on mean vs median and say why.

Part F · In-Class Quiz I simulation (20 marks · 30 minutes)

Closed book, calculator allowed. Time yourself before opening the answers file.

F1 3-1 CLO1-A

(a) **parameter**: all graduates form the population. (b) **statistic**: a sample of 60. (c) **statistic**: a sample proportion, $\hat{p}$.

F2 3-1 CLO2-A · CLO3-C

$\bar{x}$ = 152/8 = **19**; median = (15 + 17)/2 = **16**; mode = **15**. The **median** best describes a typical day: the single 40-email day pulls the mean up, and the data are right-skewed (mode < median < mean).

F3 3-2 CLO2-A

$\Sigma X = 55$, $\Sigma X^2 = 675$. $s^2 = [5(675) - 55^2]/(5 \cdot 4) = (3375 - 3025)/20 = 17.5$, so s = **4.183**. $\bar{x} = 11$, so CV = **38.03%**.

F4 3-2 CLO2-A

k = (157.5 − 120)/15 = 2.5, so at least 1 − 1/6.25 = 84%. 0.84 × 80 = 67.2, rounded up: at least **68 commutes**.

F5 3-3 CLO3-C

$z_{Stat} = (72 - 64)/5 = 1.6$ and $z_{Calc} = (81 - 75)/4 = 1.5$: **Statistics**, though the margin is small.

F6 3-4 CLO3-B

The median (70) is much closer to Q_3 (5 below) than to Q_1 (25 above), and the left whisker (25) is far longer than the right (5): **negatively skewed**. At most 45 is at or below Q_1: **25%**.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · **07 · Full worked answers**